# GAURAV KUMAR

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## PROFESSIONAL SUMMARY

Results-driven Full Stack Developer with 4+ years of experience architecting and delivering Java/Spring Boot microservices and enterprise Angular applications at scale. Deep end-to-end ownership across backend services, REST APIs, business logic, database integration, responsive UI, testing, production support, and enterprise application delivery. Led Java modernization initiatives, including Java 11→17 and 17→21 migrations, Spring 6/Spring Boot 3 upgrades, multithreading, synchronization, and concurrency fixes, performance tuning, and distributed-system troubleshooting. Early adopter of AI-assisted software development using GitHub Copilot, Amazon Q, Kiro, AI-DLC, and SpecKit, building reusable Skills, Instructions, and Agents to accelerate code analysis, migration, refactoring, and defect resolution.

## KEY ACHIEVEMENTS

- Boosted PostgreSQL throughput by 25% by re-architecting database connectivity (C3P0 → HikariCP)
- Designed and built an Agentic AI framework (Spring AI) to automate IRS Form 1095-C compliance validation
- Led greenfield development of an enterprise onboarding platform, replacing a fully manual process with automated Workday/Mosaic integrations
- Drove Java 11→17→21 and Spring Boot 3 modernization using AI-assisted tooling (Copilot, Amazon Q, Kiro), resolving critical multithreading and concurrency defects

## TECHNICAL EXPERTISE

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| **Microservices** | AWS Cloud Map (Service Discovery), Amazon API Gateway & Application Load Balancer, Feign-Client, Resilience4j (Circuit Breaker), Load Balancing, Kafka |
| **Backend** | Core Java 11/17/21, Spring, Spring Boot 3, Spring MVC, Spring Security, Spring Data JPA, Hibernate 6, HikariCP, RESTful APIs, OAuth2/JWT, Multithreading, Concurrency, Synchronization, Java I/O/NIO, SonarQube |
| **Frontend** | Angular 14/21, Node.js 14/22, TypeScript, JavaScript, RxJS (Observables, Subjects, Pipeable Operators), Reactive Forms, Ag-Grid, Highcharts, SCSS/CSS, Jasmine/Karma/Jest |
| **Cloud & AI** | AWS (ECS, Lambda, EventBridge, SSM, S3, Cloud Map, API Gateway), Agentic AI, Spring AI, LLM Instructions, Prompt Engineering, GitHub Copilot, Amazon Q, Kiro, AI-DLC, SpecKit |
| **Database & Ops** | PostgreSQL, Oracle, Docker, Kubernetes, Jenkins, CI/CD, Git, Splunk, JUnit, Mockito, TDD |
| **Software Design** | Microservices Architecture, System Design, Design Patterns, SOLID Principles, OOP, API Design, Event-Driven Architecture, Scalability & Fault Tolerance |

## PROFESSIONAL EXPERIENCE

### Senior Software Developer | ADP Private Limited

*July 2022 – Present*

***Health Compliance & Distributed Systems Modernization (ACA)***

- **Scalable Microservices:** Architected and managed a distributed microservices ecosystem on AWS ECS, using AWS Cloud Map for dynamic service discovery and Amazon API Gateway/ALB for centralized request routing and rate limiting.
- **AI-Assisted Modernization:** Leveraged GitHub Copilot, Amazon Q, Kiro, AI-DLC, and SpecKit to drive Java 11→17 and 17→21 and Spring/Spring Boot 3 modernization — cutting manual effort in code analysis, refactoring, and dependency upgrades.
- **Concurrency & Multithreading Fixes:** Diagnosed and resolved critical multithreading, synchronization, and concurrency defects in distributed services, improving system stability and eliminating race-condition-driven production issues.
- **Performance Engineering:** Re-architected database connectivity (C3P0 → HikariCP), cutting connection overhead and boosting PostgreSQL throughput by 25%.
- **Production Support & RCA:** Owned production/QA issue resolution end-to-end — tracing failures across UI, REST APIs, services, and logs, driving root cause analysis, and delivering permanent fixes that reduced recurrence.

- **Full-Stack UI Development:** Built enterprise Angular screens integrated with Spring Boot REST APIs, using Reactive Forms, reusable components, and RxJS-based async workflows.
- **Data Visualization:** Delivered interactive dashboards using ag-Grid Master-Detail and Highcharts, optimized for rendering performance and usability.
- **Event-Driven Architecture:** Used AWS EventBridge and Kafka to decouple high-volume compliance processing services, ensuring resilience during peak loads.
- **AI Engineering:** Designed an Agentic AI framework on Spring AI to automate IRS Form 1095-C data validation using structured LLM Instructions and Prompt Engineering; built reusable Skills, Instructions, and Agents to standardize AI-assisted workflows across the team.
- **Technical Leadership:** Took ownership of complex stories and production issues, mentored junior developers, drove sprint planning and code reviews, and collaborated cross-functionally to deliver on schedule.

***New Hire Reporting (NHR / PRWORA) – Greenfield Application***

- **Full-Stack Ownership:** Led greenfield development of an enterprise onboarding application, integrating Spring Boot 3 microservices with a reactive Angular frontend — from architecture through delivery.
- **Automated Workday Client Onboarding:** Architected an end-to-end automated onboarding platform, replacing a fully manual process — integrating Spring Boot 3 with Workday, Mosaic, and CDN via Feign-Client (sync) and Kafka (async) for data aggregation and workflow orchestration.
- **Reporting & Billing Automation:** Built automated JasperReports-based monthly PDF reporting and a scheduled billing batch job with automated email notifications — eliminating manual report creation and billing bottlenecks with built-in error handling and retry logic.
- **Cloud Infrastructure:** Used AWS SSM for centralized secret/config management; deployed and maintained services on AWS ECS across Dev, QA, and Prod.
- **System Design:** Designed the end-to-end architecture for the NHR onboarding platform, defining service boundaries and sync (Feign-Client) vs. async (Kafka) data flows across Workday, Mosaic, and CDN integrations to balance consistency, scalability, and fault tolerance.

**Trainee Software Engineer | Calsoft Inc** *Jan 2022 – May 2022*

- Developed a custom UI for NetApp testing infrastructure using Angular and TypeScript, automating report generation and QA verification.
- Collaborated in an Agile/Scrum environment, ensuring high code quality through JUnit/Mockito unit testing.

## EDUCATION

**B.Tech in Computer Science Engineering** | JNTU Kakinada (2018 – 2022)

**Senior Secondary (PCM)** | St. Kabir College, Patna (2018)